Module 01: Systems in Embodied Cognition — Moving Through World

Learning Objectives

1. Define the **embodied system-in-environment** as the fundamental unit of analysis for movement through the world — not the body *or* the environment, but their coupled dynamical system.
2. Analyze how the **body-environment coupling** creates a system with emergent properties not reducible to either component alone.
3. Apply the Active Inference framework to explain **locomotion, wayfinding, and spatial orientation** as system-level inference.

Introduction

We do not move through the world like objects in a void — we move *with* the world, coupled to it through every footstep, every breath of wind, every change in terrain. The system of interest in embodied movement is not the body alone but the **body-environment dyad**: a coupled dynamical system in which the body's actions change the environment and the environment's feedback changes the body.

A runner on a trail is a system: the terrain surface (roots, rocks, mud), gravitational field, air resistance, and ambient light conditions are as much part of the movement system as the runner's legs, lungs, and balance organs. Understanding movement through the world requires analyzing the whole system.

Key Concepts

1. The Body-Environment System as Coupled Dynamical System

In Active Inference, the body-environment system has a clear formalization:

- **The body** maintains internal states (proprioceptive signals, interoceptive state, motor plans) separated from the environment by a Markov blanket (the skin, the retina, the cochlea)
- **The environment** has its own dynamics (gravity, terrain, obstacles, weather) that evolve independently but are coupled to the body through the blanket
- **The coupling** is bidirectional: the body's *actions* change the environment (footfalls compress soil, hands open doors, voice alters social dynamics), and the environment's *signals* change the body's observations

The system's behavior — walking, running, climbing, swimming — is emergent: it cannot be understood by studying the body or the environment in isolation.

2. Affordance Landscapes

Gibson's affordances become dynamic when the agent moves through the world:

- **A flat surface affords walking** — but only if the surface is within reach, the agent is upright, and the agent's motor generative model includes walking policies
- **A gap affords jumping** — but only if the gap width is within the agent's estimated jump distance (calibrated by body size, fatigue, and surface traction)
- **An affordance landscape** is the spatial distribution of action possibilities as perceived by the moving agent — it shifts continuously as the agent traverses the environment

In Active Inference terms, the affordance landscape is the **spatial distribution of Expected Free Energy** across possible actions at each location. The agent moves toward regions of low EFE (reachable goals, navigable paths) and away from regions of high EFE (hazards, dead ends, extreme uncertainty).

3. Wayfinding as Active Inference

Navigation through the world is inference under a spatial generative model:

- **Cognitive map**: The agent's internal model of spatial layout (place cells, grid cells, or their functional analogs)
- **Path integration**: Continuous updating of the estimated position based on self-motion signals (proprioceptive, vestibular, optic flow)
- **Landmark recognition**: Observation-driven correction of the position estimate (analogous to the Kalman filter's update step)
- **Active exploration**: The agent selects routes that balance pragmatic value (reaching the goal) with epistemic value (exploring unknown regions to improve the map)

4. Environmental Coupling and Terrain Adaptation

Terrain adaptation demonstrates the body-environment coupling in action:

- On loose gravel: the agent reduces stride length, increases ground contact time, and lowers center of mass — precision on vestibular prediction errors increases to maintain balance
- On ice: the agent shifts to a penguin-like wide stance, drastically reducing stride length — the proprioceptive model adapts to the reduced friction by predicting smaller force vectors
- On steep inclines: the agent engages different muscle synergies (quadriceps for uphill, eccentric hamstring for downhill) — the motor generative model selects slope-appropriate policies

Applications

- **Wilderness navigation training**: Teaching backcountry navigation through active inference involves calibrating the proprioceptive model to terrain signals — experienced hikers can estimate distance traveled from step count adjusted for slope and surface type, implementing path integration through somatic channels.
- **Urban accessibility design**: Designing cities for wheelchair users requires understanding the affordance landscape from the wheelchair user's perspective — curb

heights, surface textures, and slope gradients define the distribution of navigable vs. impassable regions, and optimal accessible design minimizes the free energy of movement for all body configurations.

Conclusion

Moving through the world is system-level inference: the body-environment coupling, affordance landscapes, wayfinding, and terrain adaptation are all manifestations of Active Inference in the dynamical system formed by the agent and its milieu. The next module examines the embodied agent within this system.